

MTH-262 Statistics and Probability Theory

Lecture 12



Review of Lecture 11

In the last lecture, we discussed discrete and continuous random variables in detail.

Lecture 11:

- Discrete random variable and its probability distribution.
- Histogram of probability distribution.
- Cumulative distribution and its graph.
- Continuous random variable and its probability distribution.
- Cumulative distribution function and its graph.

Joint Probability Distribution

- Our study of random variables and their probability distributions in the preceding sections is restricted to one-dimensional sample spaces, in that we recorded outcomes of an experiment as values assumed by a single random variable.
- There will be situations, however, where we may find it desirable to record the simultaneous outcomes of several random variables.
- For example, we might measure the amount of precipitate P and volume V of gas released from a controlled chemical experiment, giving rise to a two-dimensional sample space consisting of the outcomes (p, v) ,
- We might be interested in the hardness H and tensile strength T of cold-drawn copper, resulting in the outcomes (h, t) .

Joint Probability Distribution

- In a study to determine the likelihood of success in college based on high school data, we might use a three dimensional sample space and record for each individual his or her aptitude test score, high school class rank, and grade-point average at the end of freshman year in college.
- If X and Y are two discrete random variables, the probability distribution for their simultaneous occurrence can be represented by a function with values $f(x, y)$ for any pair of values (x, y) within the range of the random variables X and Y .
- It is customary to refer to this function as the joint probability distribution of X and Y . Hence, in the discrete case,

$$f(x, y) = P(X = x, Y = y);$$

Joint Probability Distribution of discrete random variable

The function $f(x, y)$ is a joint probability distribution or probability mass function of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y) ,
2. $\sum_x \sum_y f(x, y) = 1$
3. $P(X = x, Y = y) = f(x, y)$.

For any region A in the xy plane, $P[(X, Y) \in A] = \sum \sum_A f(x, y)$.

Joint Probability Distribution (Example)

Question: Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If $\underline{X}$ is the number of blue pens selected and $\underline{Y}$ is the number of red pens selected, find

(a) the joint probability function $f(x, y)$,

(b) $P[(X, Y) \in A]$, where A is the region $\{(x, y) \mid x + y \leq 1\}$.

Joint Probability Distribution (Example)

Solution: The possible pairs of values (x, y) are $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, $(0, 2)$, and $(2, 0)$.

(a) Now, $f(1, 1)$, for example, represents the probability that a red and a green pens are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is $\binom{8}{2} = 28$.

The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $\binom{2}{1}\binom{3}{1} = 6$. Hence, $f(1, 1) = 6/28 = 3/14$. Similar calculations yield the probabilities for the other cases, which are presented in Table (See next slide). Note that the probabilities sum to 1.

Joint Probability Distribution (Example)

All corresponding probabilities are given in the following table:

Joint Probability Distribution

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Joint Probability Distribution (Example)

(a) Joint Probability function:

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}},$$

for $x = 0, 1, 2$; $y = 0, 1, 2$; and $0 \leq x + y \leq 2$.

Joint Probability Distribution

(b) The probability that (X, Y) fall in the region A is:

$$\begin{aligned} P[(X, Y) \in A] &= P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0) \\ &= \frac{3}{28} + \frac{9}{28} + \frac{3}{14} = \frac{9}{14} \end{aligned}$$

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Joint Probability Distribution of Continuous Random Variables

The function $f(x, y)$ is a joint density function of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y) ,
2. $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$
3. $P[(X, Y) \in A] = \iint_A f(x, y) = dx dy$ for any region A in the xy plane.

Joint Probability Distribution (Example)

Question: A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (i) Verify condition 2 of Definition.
- (ii) Find $P[(X, Y) \in A]$, where $A = \{(x, y) \mid 0 < x < 1/2, 1/4 < y < 1/2\}$.

Joint Probability Distribution (Example)

(a) The integration of $f(x, y)$ over the whole region is

$$\begin{aligned}\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy &= \int_0^1 \int_0^1 \frac{2}{5} (2x + 3y) \, dx \, dy \\ &= \int_0^1 \left(\frac{2x^2}{5} + \frac{6xy}{5} \right) \Big|_{x=0}^{x=1} dy \\ &= \int_0^1 \left(\frac{2}{5} + \frac{6y}{5} \right) dy = \left(\frac{2y}{5} + \frac{3y^2}{5} \right) \Big|_0^1 = \frac{2}{5} + \frac{3}{5} = 1.\end{aligned}$$

Joint Probability Distribution (Example)

(b) To calculate the probability, we use

$$\begin{aligned} P[(X, Y) \in A] &= P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right) \\ &= \int_{1/4}^{1/2} \int_0^{1/2} \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_{1/4}^{1/2} \left(\frac{2x^2}{5} + \frac{6xy}{5}\right) \Big|_{x=0}^{x=1/2} dy = \int_{1/4}^{1/2} \left(\frac{1}{10} + \frac{3y}{5}\right) dy \\ &= \left(\frac{y}{10} + \frac{3y^2}{10}\right) \Big|_{1/4}^{1/2} \\ &= \frac{1}{10} \left[\left(\frac{1}{2} + \frac{3}{4}\right) - \left(\frac{1}{4} + \frac{3}{16}\right)\right] = \frac{13}{160}. \end{aligned}$$

Joint Probability Distribution (Example)

- Question: Suppose that the shelf life, in years, of a certain perishable food product packaged in cardboard containers is a random variable whose probability density function is given by:

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

- Let X_1 , X_2 , and X_3 represent the shelf lives for three of these containers selected independently and find $P(X_1 < 2, 1 < X_2 < 3, X_3 > 2)$.

Joint Probability Distribution (Example)

- Solution:

Since the containers were selected independently, we can assume that the random variables X_1 , X_2 , and X_3 are statistically independent, having the joint probability density

$$f(x_1, x_2, x_3) = f(x_1)f(x_2)f(x_3) = e^{-x_1}e^{-x_2}e^{-x_3} = e^{-x_1-x_2-x_3},$$

for $x_1 > 0$, $x_2 > 0$, $x_3 > 0$, and $f(x_1, x_2, x_3) = 0$ elsewhere. Hence

$$\begin{aligned} P(X_1 < 2, 1 < X_2 < 3, X_3 > 2) &= \int_2^\infty \int_1^3 \int_0^2 e^{-x_1-x_2-x_3} dx_1 dx_2 dx_3 \\ &= (1 - e^{-2})(e^{-1} - e^{-3})e^{-2} = 0.0372. \end{aligned}$$

Marginal Distributions

- Given the joint probability distribution $f(x, y)$ of the discrete random variables X and Y , the probability distribution $g(x)$ of X alone is obtained by summing $f(x, y)$ over the values of Y . Similarly, the probability distribution $h(y)$ of Y alone is obtained by summing $f(x, y)$ over the values of X .
- We define $g(x)$ and $h(y)$ to be the marginal distributions of X and Y , respectively. When X and Y are continuous random variables, summations are replaced by integrals. We can now make the following general definition.

The marginal distributions of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.

Marginal Distribution of discrete random variable (Example)

Question: Show that the column and row totals of following Table give the marginal distribution of X alone and of Y alone.

Joint probability distribution

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Marginal Distribution for discrete random variable (Example)

Solution: For the random variable X , we see that

$$g(0) = f(0,0) + f(0,1) + f(0,2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1,0) + f(1,1) + f(1,2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2,0) + f(2,1) + f(2,2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table. In a similar manner we could show that the values of $h(y)$ are given by the row totals. In tabular form, these marginal distributions may be written as follows:

x	0	1	2
$g(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

y	0	1	2
$h(y)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$

Marginal Distribution for continuous random variable (Example)

Question: A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $g(x)$ and $h(y)$ for the joint density function of drive-in and walk-in facilities.

Marginal Distribution for continuous random variable (Example)

Solution: By definition

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy = \int_0^1 \frac{2}{5}(2x + 3y) \, dy = \left(\frac{4xy}{5} + \frac{6y^2}{10} \right) \Big|_{y=0}^{y=1} = \frac{4x + 3}{5},$$

for $0 \leq x \leq 1$, and $g(x) = 0$ elsewhere.

Similarly

$$h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^1 \frac{2}{5}(2x + 3y) \, dx = \frac{2(1 + 3y)}{5},$$

for $0 \leq y \leq 1$, and $h(y) = 0$ elsewhere.

$g(X)$ and $h(Y)$ are the marginal distribution of drive-in and walk-in facilities.